

REQUIREMENTS TRACEABILITY MATRIX

CSI-100 Cell Sorting Instrument (IVD)

Document number	TM-CSI-DT-01
Revision	C
Document type	Traceability matrix (design history file)
Status	Released
Status date	14 May 2027 (V&V complete)
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Product	CSI-100 Cell Sorting Instrument, all configurations

Purpose

Bidirectional traceability from user needs through system requirements, software requirements, design outputs, verification tests and validation evidence, with risk controls, anomalies and production controls. Generated from the controlled program documents listed in Section 1.3.

Document Control

Rev	Date	Description of change	Author
A	Nov 2025	Initial matrix: user needs, system requirements, allocation.	A. Lodhi
B	Aug 2026	Added software requirements, design outputs, verification protocols and design transfer links (DTRR-1).	A. Lodhi
C	May 2027	V&V complete: verification and validation results, anomaly links, software architecture allocation, backward trace and completeness checks. Baseline for VVR-CSI-001 and the 510(k) submission.	A. Lodhi

Approvals

Function	Role	Signature / Date
Systems Engineering	Author, traceability owner	
Software Engineering	Software traceability	
V&V Engineering	Verification results	
Clinical Affairs	Validation results	
Quality Engineering	Design history file, risk management	
Regulatory Affairs	Submission	

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1. Introduction

1.1 Purpose

This matrix demonstrates that the CSI-100 design control chain is complete. It links every user need to the system requirements that address it and the validation evidence that confirms it, every system requirement to its design outputs and verification, and every software requirement to its parent, its software item and its tests. It also traces backward, so that every test and every design element is justified by a requirement.

1.2 Trace model

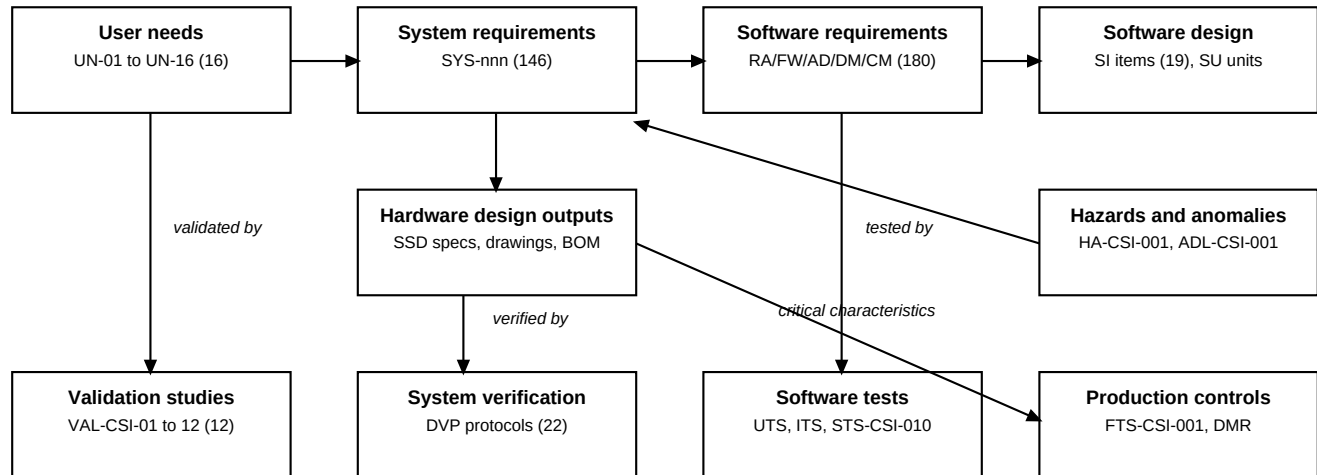


Figure 1. Trace model. Every arrow is traced in both directions in this document.

1.3 Source documents and baseline

Trace element	Source document (baseline)
User needs, system requirements, source, allocation, verification method	SRS-CSI-001 Rev D
Software requirements, parents, items, test level	SRS-SW-CSI-010 Rev E
Software items and units	SAD-SW-CSI-011 Rev C
Hardware design outputs	SSD-CSI-100 to SSD-CSI-950, drawings, BOM (DTP-CSI-001 Section 16)
Verification protocols, validation studies, risk control verification	VVP-CSI-001 Rev A
Verification and validation results	PVR-CSI-001, SVR-CSI-001, SVR-SW-CSI-010, DVAL-CSI-001
Anomalies	ADL-CSI-001 Export 07
Production controls	DTP-CSI-001 Rev B, FTS-CSI-001 Rev A

The matrix is exported from the requirements management tool. For this revision every table and every completeness check in Section 2 was generated from the source documents above, so the counts and links reflect those documents exactly. Result 'Pass' means the requirement met its acceptance criterion in the cited report; 'after ANM-nnnn' means it passed after the listed anomaly was corrected and re-verified.

2. Completeness Summary

These checks answer the two questions auditors ask first: is every requirement tested, and is every test justified by a requirement? Each check is computed from the matrices in this document.

Check	Items	Traced	Result
Every user need traces to at least one system requirement	16	16	Complete
Every user need traces to at least one validation study	16	16	Complete
Every system requirement has a source (user need, hazard, standard or PRD)	146	146	Complete
Every system requirement is allocated to a subsystem or the system	146	146	Complete
Every system requirement has a design output	146	146	Complete
Every system requirement has a verification method and protocol	146	146	Complete
Every system requirement verified with result Pass	146	146	Complete
Every software requirement has a valid parent (SYS or HA)	180	180	Complete
Every software requirement is allocated to a software item in SAD-SW-CSI-011	180	180	Complete
Every software requirement has a test at its test level	180	180	Complete
Every system requirement allocated to software (SS-10) has a child software requirement	38	38	Complete
Every verification protocol traces to at least one system requirement (no unjustified tests)	22	22	Complete
Every validation study traces to at least one user need	12	12	Complete
Every software item in the architecture implements at least one software requirement	19	19	Complete
Every anomaly traces to a requirement or user need	32	32	Complete
Every hazard with a design risk control traces to verification evidence	14	14	Complete

Conclusion: all traceability checks are complete; no orphan requirements, tests, design items or validation studies. Open or deferred anomalies (ANM-0053, ANM-0062, ANM-0064, ANM-0066) are all severity 3 or 4 and do not change any verification result (ADL-CSI-001 Section 9).

Level	Count
User needs	16
System requirements	146
System requirements allocated to software (SS-10)	38
Software requirements	180
Software items / units	19 / 97
System verification protocols	22
Validation studies	12
Hazards with design risk controls	14
Anomalies linked	32

3. Matrix A: User Needs to System Requirements and Validation

Forward trace from each user need to the system requirements that satisfy it (verified in Matrix B) and to the validation studies that confirm it. Verification and validation are shown in separate columns and are never used as evidence for each other.

User need	Statement	System requirements	Validation	Validation result
UN-01	Identify and enumerate cell populations in stained human specimens with results accurate ...	401, 402, 411, 421, 422, 433, 434, 443, 444, 445, 495, 632, 704, 806	VAL-01, VAL-04	Pass
UN-02	Physically separate target cell populations at high purity and recovery, with cells ...	461, 462, 464, 466, 631, 633, 634	VAL-02	Pass (after ANM-0056)
UN-03	Measure many markers on each cell at the same time (multicolor panels).	431, 432, 435, 441, 632	VAL-01	Pass
UN-04	Process specimens at a throughput that fits the laboratory workday.	451, 452, 453, 492, 494, 604	VAL-05	Pass
UN-05	Collect sorted cells into tubes, multiwell plates and slides, including single-cell ...	464, 481, 482, 483, 484, 485, 631	VAL-02	Pass (after ANM-0056)
UN-06	Operate the instrument without risk of injury from laser light, high voltage, moving ...	551, 803, 804, 902, 1304	VAL-03	Pass (after ANM-0060)
UN-07	Install the instrument in a standard clinical laboratory without special facilities.	301, 302, 303, 304, 305, 306, 311, 312, 313, 321, 322, 324, 325, 326, 331, 438, 1104, 1201, 1203, 1204	VAL-06	Pass
UN-08	Obtain consistent performance from day to day with minimal user adjustment, confirmed by ...	436, 437, 444, 465, 468, 613, 621, 622, 624, 806, 1201	VAL-04	Pass
UN-09	Set up, run, clean and shut down the instrument through guided, largely automated ...	437, 465, 492, 493, 601, 602, 603, 611, 612, 614, 805, 901	VAL-03	Pass (after ANM-0060)
UN-10	Keep results and records secure, attributable and traceable, with controlled user access.	635, 701, 702, 703, 705, 706, 709, 802, 904, 1007	VAL-09	Pass (after ANM-0058)
UN-11	Keep samples and sorted cells at a chosen temperature during the run.	4101, 4102, 803	VAL-08	Pass
UN-12	Keep the instrument in service with predictable maintenance and fast repair.	1001, 1002, 1003, 1004, 1005, 1006, 1007, 1104, 1202	VAL-10	Pass
UN-13	Use the instrument in the United States and the European Union in compliance with IVD ...	1101, 1102, 1105, 1301, 1302, 1303, 1304, 1305, 1306	VAL-12	Pass
UN-14	Avoid carryover and cross-contamination between specimens.	471, 472, 603	VAL-07	Pass
UN-15	Handle cells of different sizes and fragility.	423, 463, 467, 491, 611	VAL-02	Pass (after ANM-0056)
UN-16	Choose the laser and detector configuration that fits the laboratory's assays, and ...	431, 432, 442, 623, 1008	VAL-11	Pass

System requirement numbers shown without the SYS- prefix. Validation results from DVAL-CSI-001.

4. Matrix B: System Requirements to Design Outputs and Verification

One row per system requirement in SRS-CSI-001. Design output lists the subsystem specification and drawings for hardware allocations, and the child software requirements for software allocations. Requirement text is abbreviated; the SRS is the controlling text.

ID	Requirement (abbreviated)	Source	Design output	Verification	Result
SYS-301	The sorter shall not exceed 71 cm height, 122 cm width and 71 cm depth (28 × 48 × 28 in) ...	UN-07	DWG-CSI-0001; BOM-CSI-0001	I: DVP-301; PVR-CSI-001	Pass
SYS-302	The sorter shall weigh no more than 185 kg (408 lb) in its heaviest configuration.	UN-07	DWG-CSI-0001; BOM-CSI-0001	I: DVP-301; PVR-CSI-001	Pass
SYS-303	The fluidics supply module shall not exceed 66 cm height, 81 cm width and 66 cm depth (26 × 32 ...	UN-07	SSD-CSI-100; DWG-CSI-1000, -1100	I: DVP-301; PVR-CSI-001	Pass
SYS-304	The fluidics supply module shall be mobile on lockable casters with leveling feet.	UN-07	SSD-CSI-100; DWG-CSI-1000, -1100	I: DVP-301; PVR-CSI-001	Pass
SYS-305	The sorter shall operate on a laboratory bench or on an optional bench no larger than 85.1 × ...	UN-07	DWG-CSI-0001; BOM-CSI-0001	I: DVP-301; PVR-CSI-001	Pass
SYS-306	The umbilical between sorter and fluidics supply module shall allow separation of up to 2.7 m ...	UN-07	SSD-CSI-100; DWG-CSI-1000, -1100	I: DVP-301; PVR-CSI-001	Pass
SYS-311	The system shall operate from 100, 115 or 230 VAC (±10%), 50 or 60 Hz, single phase.	UN-07	SSD-CSI-600; SCH-CSI-6000	T: DVP-301; PVR-CSI-001	Pass
SYS-312	Total power consumption of the sorter and fluidics supply module shall not exceed 1,500 W.	UN-07	DWG-CSI-0001; BOM-CSI-0001	T: DVP-301; PVR-CSI-001	Pass
SYS-313	The system shall require no more than one 20 A outlet for the sorter and fluidics supply ...	UN-07	DWG-CSI-0001; BOM-CSI-0001	A: DVP-301; PVR-CSI-001	Pass
SYS-314	The fluidics supply module shall be powered and switched only from the sorter.	HA-12	SSD-CSI-100; DWG-CSI-1000, -1100; SSD-CSI-600; SCH-CSI-6000	I: DVP-301; PVR-CSI-001	Pass
SYS-315	The system shall be protected by a resettable circuit breaker.	HA-12	SSD-CSI-600; SCH-CSI-6000	I: DVP-301; PVR-CSI-001	Pass
SYS-316	The system shall meet installation category II and pollution degree 2 as defined in IEC ...	IEC 61010-1	DWG-CSI-0001; BOM-CSI-0001	A: DVP-301; PVR-CSI-001	Pass
SYS-321	The system shall meet all performance requirements at ambient temperatures of 15 to 30 °C.	UN-07	DWG-CSI-0001; BOM-CSI-0001	T: DVP-301; PVR-CSI-001	Pass
SYS-322	The system shall meet all performance requirements at 5 to 80% relative humidity ...	UN-07	DWG-CSI-0001; BOM-CSI-0001	T: DVP-301; PVR-CSI-001	Pass
SYS-323	The system shall operate at altitudes up to 2,000 m.	IEC 61010-1	DWG-CSI-0001; BOM-CSI-0001	A: DVP-301; PVR-CSI-001	Pass

ID	Requirement (abbreviated)	Source	Design output	Verification	Result
SYS-324	The system shall withstand storage at 1 to 40 °C without loss of performance.	UN-07	DWG-CSI-0001; BOM-CSI-0001	T: DVP-301; PVR-CSI-001	Pass
SYS-325	The system shall require no special room facilities, no external compressed air and no ...	UN-07	SSD-CSI-100; DWG-CSI-1000, -1100	D: DVP-301; PVR-CSI-001	Pass
SYS-326	The system shall accept optional facility air at 80 to 100 PSI, dry and oil-free.	UN-07	SSD-CSI-100; DWG-CSI-1000, -1100	T: DVP-301; PVR-CSI-001	Pass
SYS-331	Acoustic noise at the operator position shall not exceed 62 dBA during sorting.	UN-07	DWG-CSI-0001; BOM-CSI-0001	T: DVP-301; PVR-CSI-001	Pass
SYS-401	The coefficient of variation (CV) of the full G0/G1 peak for propidium iodide-stained chicken ...	UN-01	SSD-CSI-300; DWG-CSI-3100; SSD-CSI-400; DWG-CSI-4000; SSD-CSI-500; DWG-CSI-5100, -5200; SRS-SW: AD-304	T: DVP-401; PVR-CSI-001	Pass (after ANM-0012)
SYS-402	The CV of the full G0/G1 peak for Hoechst-stained chicken erythrocyte nuclei, excited at 405 ...	UN-01	SSD-CSI-300; DWG-CSI-3100; SSD-CSI-400; DWG-CSI-4000; SSD-CSI-500; DWG-CSI-5100, -5200; SRS-SW: AD-304	T: DVP-401; PVR-CSI-001	Pass
SYS-411	The doublet-to-singlet ratio for chicken erythrocyte nuclei shall be 1.95 to 2.05, for PI ...	UN-01	SSD-CSI-500; DWG-CSI-5100, -5200; SSD-CSI-600; SCH-CSI-6000; SRS-SW: AD-303, AD-304	T: DVP-401; PVR-CSI-001	Pass
SYS-421	Forward and side scatter shall resolve 0.5 µm beads from instrument noise.	UN-01	SSD-CSI-400; DWG-CSI-4000; SSD-CSI-500; DWG-CSI-5100, -5200	T: DVP-402; PVR-CSI-001	Pass
SYS-422	Scatter sensitivity shall allow separation of fixed platelets from noise and identification of ...	UN-01	SSD-CSI-400; DWG-CSI-4000; SSD-CSI-500; DWG-CSI-5100, -5200	T: DVP-402; PVR-CSI-001	Pass
SYS-423	The forward scatter path shall accept interchangeable neutral density filters of 0.5, 1.0, 1.5 ...	UN-15	SSD-CSI-500; DWG-CSI-5100, -5200; SRS-SW: AD-305	I: DVP-402; PVR-CSI-001	Pass
SYS-431	The system shall support up to six installed lasers, with up to four active at the same time.	UN-03, UN-16	SSD-CSI-400; DWG-CSI-4000	D: DVP-403; PVR-CSI-001	Pass
SYS-432	The 488 nm and 633 nm lasers shall be standard in every configuration. 375, 405, 445 and 561 ...	UN-03, UN-16	SSD-CSI-400; DWG-CSI-4000	I: DVP-403; PVR-CSI-001	Pass
SYS-433	Minimum laser power out of the laser head shall be: 375 nm > 7 mW; 405 nm > 50 mW; 445 nm > 20 ...	UN-01	SSD-CSI-400; DWG-CSI-4000; SRS-SW: FW-305, AD-201, AD-202, AD-203	T: DVP-403; PVR-CSI-001	Pass

ID	Requirement (abbreviated)	Source	Design output	Verification	Result
SYS-434	Elliptical beams at the interrogation point shall measure $9 \pm 3 \mu\text{m}$ in height and $65 \pm 7 \mu\text{m}$ in ...	UN-01	SSD-CSI-400; DWG-CSI-4000	T: DVP-403; PVR-CSI-001	Pass
SYS-435	Each active laser shall be focused on a spatially separated beam spot. Lasers sharing a spot ...	UN-03	SSD-CSI-400; DWG-CSI-4000; SSD-CSI-600; SCH-CSI-6000; SRS-SW: FW-305, FW-306	T: DVP-403; PVR-CSI-001	Pass
SYS-436	Optical alignment shall be fixed at the factory and shall require no user adjustment.	UN-08	SSD-CSI-400; DWG-CSI-4000; SRS-SW: AD-301	D: DVP-403; PVR-CSI-001	Pass
SYS-437	The system shall reach full performance within 30 minutes of laser power-on.	UN-08, UN-09	SSD-CSI-400; DWG-CSI-4000; SRS-SW: AD-204	T: DVP-403; PVR-CSI-001	Pass
SYS-438	Lasers shall be air-cooled and shall need no external cooling or special power.	UN-07	SSD-CSI-400; DWG-CSI-4000	I: DVP-403; PVR-CSI-001	Pass
SYS-441	The system shall measure forward scatter, side scatter and at least 18 fluorescence parameters ...	UN-03	SSD-CSI-500; DWG-CSI-5100, -5200; SRS-SW: FW-304, AD-302	T: DVP-402; PVR-CSI-001	Pass
SYS-442	Each fluorescence detector shall have a user-replaceable bandpass filter and each detector ...	UN-16	SSD-CSI-500; DWG-CSI-5100, -5200	I: DVP-402; PVR-CSI-001	Pass
SYS-443	The system shall report height, area and width for every detected parameter.	UN-01	SSD-CSI-600; SCH-CSI-6000; SRS-SW: RA-301, FW-201, FW-303	T: DVP-402; PVR-CSI-001	Pass
SYS-444	Signals from each beam spot shall be time-aligned to the same event, with inter-laser delay ...	UN-01, UN-08	SSD-CSI-600; SCH-CSI-6000; SRS-SW: RA-802, FW-201, AD-205	T: DVP-402; PVR-CSI-001	Pass
SYS-445	Data shall be digitized with a linear dynamic range of at least 18 bits.	UN-01	SSD-CSI-600; SCH-CSI-6000; SRS-SW: RA-302, FW-202	A: DVP-402; PVR-CSI-001	Pass
SYS-451	The system shall acquire at least 70,000 events per second with 8 parameters and 12 ...	UN-04	SSD-CSI-600; SCH-CSI-6000; SRS-SW: RA-303, RA-304, FW-202, FW-806	T: DVP-404; PVR-CSI-001	Pass
SYS-452	The system shall acquire more than 100,000 events per second when fewer parameters are ...	UN-04	SSD-CSI-600; SCH-CSI-6000; SRS-SW: RA-303, FW-202, FW-806	T: DVP-404; PVR-CSI-001	Pass
SYS-453	Sample flow rate shall be user-selectable over at least 10 to 80 $\mu\text{L}/\text{min}$.	UN-04	SSD-CSI-100; SRS-SW: RA-206, AD-402	T: DVP-404; PVR-CSI-001	Pass
SYS-461	For a four-way sort at 70 PSI, 90 kHz drop drive and an average threshold rate of 25,000 ...	UN-02	SSD-CSI-600; SCH-CSI-6000; SSD-CSI-700; DWG-CSI-7000; SRS-SW: FW-204	T: DVP-405; PVR-CSI-001	Pass (after ANM-0003, ANM-0047)
SYS-462	The system shall sort into up to four populations at the same time.	UN-02	SSD-CSI-700; DWG-CSI-7000	D: DVP-405; PVR-CSI-001	Pass

ID	Requirement (abbreviated)	Source	Design output	Verification	Result
SYS-463	Drop drive frequency shall be adjustable from 1 to 100 kHz or wider.	UN-15	SSD-CSI-700; DWG-CSI-7000; SRS-SW: RA-405, AD-502	T: DVP-405; PVR-CSI-001	Pass
SYS-464	The system shall resolve particle position within the drop to 1/32 of a drop and shall provide ...	UN-02, UN-05	SSD-CSI-600; SCH-CSI-6000; SRS-SW: RA-604, RA-605, FW-203, AD-703	T: DVP-405; PVR-CSI-001	Pass
SYS-465	The system shall automatically determine drop delay using fluorescent drop-delay beads.	UN-08, UN-09	SSD-CSI-700; DWG-CSI-7000; SRS-SW: RA-407	T: DVP-405; PVR-CSI-001	Pass
SYS-466	Sorted cell lines and human PBMCs shall remain viable and proliferate after sorting at every ...	UN-02	DWG-CSI-0001; BOM-CSI-0001	T: DVP-405; PVR-CSI-001	Pass
SYS-467	The system shall provide nozzles of 70, 85 and 100 µm, with a 130 µm nozzle as an option, each ...	UN-15	SSD-CSI-300; DWG-CSI-3100; SSD-CSI-700; DWG-CSI-7000	I: DVP-405; PVR-CSI-001	Pass
SYS-468	Changing the nozzle shall not require optical realignment.	UN-08	SSD-CSI-300; DWG-CSI-3100; SRS-SW: AD-503	D: DVP-405; PVR-CSI-001	Pass
SYS-471	Sample carryover from one tube to the next shall not exceed 0.5% after the automatic sample ...	UN-14	SSD-CSI-100	T: DVP-406; PVR-CSI-001	Pass (after ANM-0019)
SYS-472	The system shall flush the inside and outside of the sample line with sheath after every tube ...	UN-14	SSD-CSI-100; SRS-SW: RA-205	D: DVP-406; PVR-CSI-001	Pass
SYS-481	The system shall provide two-way sorting into 12 × 75 mm and 15 mL tubes, and four-way sorting ...	UN-05	SSD-CSI-800; SRS-SW: RA-601	D: DVP-407; PVR-CSI-001	Pass
SYS-482	With the plate deposition option, the system shall sort into 6-, 24-, 48-, 96- and 384-well ...	UN-05	SSD-CSI-800; SRS-SW: RA-601, FW-208	D: DVP-407; PVR-CSI-001	Pass
SYS-483	With the plate deposition option, the system shall deposit single cells so that at least 95% ...	UN-05	SSD-CSI-700; DWG-CSI-7000; SSD-CSI-800; SRS-SW: FW-206, FW-208, AD-504	T: DVP-407; PVR-CSI-001	Pass (after ANM-0037)
SYS-484	The system shall record the measured parameters of each cell deposited in a plate well (index ...	UN-05	SRS-SW: RA-705	D: DVP-407; PVR-CSI-001	Pass
SYS-485	The system shall allow users to define custom collection devices.	UN-05	SRS-SW: RA-601, AD-702	D: DVP-407; PVR-CSI-001	Pass
SYS-491	Sheath pressure shall be software-controlled from 5 to 75 PSI.	UN-15	SSD-CSI-100; DWG-CSI-1000, -1100; SRS-SW: FW-401, AD-401, AD-404	T: DVP-408; PVR-CSI-001	Pass
SYS-492	The fluidics supply module shall hold at least 10 L of sheath, 10 L of waste, 5 L of shutdown ...	UN-04, UN-09	SSD-CSI-100; DWG-CSI-1000, -1100	I: DVP-408; PVR-CSI-001	Pass

ID	Requirement (abbreviated)	Source	Design output	Verification	Result
SYS-493	The system shall warn the user before sheath runs out or waste is full, and shall prevent ...	UN-09, HA-15	SSD-CSI-100; DWG-CSI-1000, -1100; SRS-SW: RA-103, RA-207, FW-402, AD-403	T: DVP-408; PVR-CSI-001	Pass
SYS-494	The sample chamber shall accept 15 mL, 12 × 75 mm and 1 mL tubes.	UN-04	SSD-CSI-100; SRS-SW: RA-205	I: DVP-408; PVR-CSI-001	Pass
SYS-495	The system shall agitate the sample tube at user-selectable speeds.	UN-01	SSD-CSI-100; SRS-SW: RA-206	D: DVP-408; PVR-CSI-001	Pass
SYS-410 1	The sample chamber shall hold a user-selected temperature, reached within 45 minutes with the ...	UN-11	SSD-CSI-100; SRS-SW: RA-206, AD-405	T: DVP-408; PVR-CSI-001	Pass (after ANM-0039)
SYS-410 2	With the temperature control option, collection tubes shall be held at a user-selected ...	UN-11	SSD-CSI-800; SRS-SW: AD-505	T: DVP-408; PVR-CSI-001	Pass
SYS-501	The system shall comply with IEC 61010-1 and IEC 61010-2-101 for protection against electric ...	HA-12, IEC 61010	DWG-CSI-0001; BOM-CSI-0001	T: DVP-501; SVR-CSI-001	Pass
SYS-502	Protective earth continuity, dielectric strength and touch current shall be tested on every ...	HA-12	DWG-CSI-0001; BOM-CSI-0001	T: DVP-501; SVR-CSI-001	Pass
SYS-503	The system shall comply with IEC 61326-1 and IEC 61326-2-6 for electromagnetic compatibility.	HA-13, IEC 61326	DWG-CSI-0001; BOM-CSI-0001	T: DVP-501; SVR-CSI-001	Pass (after ANM-0021)
SYS-511	The system shall be a Class 1 laser product per IEC 60825-1 in all user-accessible conditions.	HA-01, IEC 60825-1	DWG-CSI-0001; BOM-CSI-0001	T: DVP-502; SVR-CSI-001	Pass
SYS-512	Opening the flow cell access door shall block all laser beams by a shutter before any beam is ...	HA-01	SSD-CSI-400; DWG-CSI-4000; SRS-SW: FW-502	T: DVP-502; SVR-CSI-001	Pass
SYS-513	Laser warning labels shall be placed on all service-access panels behind which Class 3B ...	HA-01	DWG-CSI-0001; BOM-CSI-0001	I: DVP-502; SVR-CSI-001	Pass (after ANM-0032)
SYS-521	The sort collection chamber door shall not open unless the sort block door is closed.	HA-05	SSD-CSI-700; DWG-CSI-7000; SRS-SW: FW-502	T: DVP-503; PVR-CSI-001	Pass
SYS-522	An illuminated warning light and a software indicator shall show whenever deflection plate ...	HA-02	SSD-CSI-700; DWG-CSI-7000; SRS-SW: RA-103, RA-1003, FW-608	T: DVP-503; PVR-CSI-001	Pass
SYS-523	Deflection plate voltage shall not exceed 12,000 V between plates and shall be insulated so ...	HA-02	SSD-CSI-700; DWG-CSI-7000; SRS-SW: FW-307, AD-502	T: DVP-503; PVR-CSI-001	Pass
SYS-524	The system shall fully depressurize within 10 seconds of main power switch-off.	HA-09	SSD-CSI-100; DWG-CSI-1000, -1100; SRS-SW: FW-403	T: DVP-503; PVR-CSI-001	Pass
SYS-531	The system shall provide an illuminated emergency stop button next to the sample loading port.	HA-06	SSD-CSI-600; SCH-CSI-6000	I: DVP-503; PVR-CSI-001	Pass
SYS-532	Pressing the emergency stop shall, within 2 seconds: unload the sample tube, stop plate stage ...	HA-02, HA-06	SSD-CSI-600; SCH-CSI-6000; SRS-SW: FW-601	T: DVP-503; PVR-CSI-001	Pass

ID	Requirement (abbreviated)	Source	Design output	Verification	Result
SYS-533	The emergency stop button shall light while the sample loading port is moving.	HA-06	SSD-CSI-600; SCH-CSI-6000; SRS-SW: FW-501	T: DVP-503; PVR-CSI-001	Pass
SYS-541	The system shall monitor the drop breakoff continuously during sorting.	HA-04, HA-07	SSD-CSI-700; DWG-CSI-7000; SRS-SW: FW-210	T: DVP-503; PVR-CSI-001	Pass
SYS-542	If the breakoff moves outside its tolerance, the system shall pause sorting until the breakoff ...	HA-07	SSD-CSI-700; DWG-CSI-7000; SRS-SW: FW-205	T: DVP-503; PVR-CSI-001	Pass
SYS-543	If a nozzle clog is detected, the system shall, within 2 seconds: stop the stream, turn off ...	HA-04	SSD-CSI-700; DWG-CSI-7000; SRS-SW: FW-602	T: DVP-503; PVR-CSI-001	Pass (after ANM-0011)
SYS-551	With the sort block and collection chamber doors closed and the aerosol evacuation option ...	HA-04, UN-06	SSD-CSI-900; SRS-SW: RA-208	T: DVP-504; PVR-CSI-001	Pass
SYS-552	The aerosol evacuation option shall provide at least 7 CFM in normal mode and at least 30 CFM ...	HA-04	SSD-CSI-900	T: DVP-504; PVR-CSI-001	Pass
SYS-553	All wetted surfaces and surfaces exposed to specimens shall withstand the decontamination ...	HA-10	DWG-CSI-0001; BOM-CSI-0001	T: DVP-504; PVR-CSI-001	Pass
SYS-554	The IFU shall describe safe handling of a clog and decontamination after sorting infectious ...	HA-04	DWG-CSI-0001; BOM-CSI-0001; SRS-SW: RA-504	I: DVP-504; PVR-CSI-001	Pass
SYS-561	Pressurized tanks shall be rated for at least 1.5 times the maximum operating pressure and ...	HA-09	SSD-CSI-100; DWG-CSI-1000, -1100	A, T: DVP-505; PVR-CSI-001	Pass
SYS-562	Moving parts (loading port, plate stage, aspirator drawer) shall not exert a pinching force ...	HA-06	SSD-CSI-100; SSD-CSI-800; SRS-SW: FW-309	T: DVP-505; PVR-CSI-001	Pass
SYS-563	The system shall remain stable when tilted 10° in any direction.	HA-11, IEC 61010-1	DWG-CSI-0001; BOM-CSI-0001	T: DVP-505; PVR-CSI-001	Pass
SYS-601	The system shall perform an automated fluidics startup that primes all fluid lines and removes ...	UN-09	SSD-CSI-100; DWG-CSI-1000, -1100; SRS-SW: RA-201	D: DVP-601; PVR-CSI-001	Pass
SYS-602	The system shall perform an automated fluidics shutdown that leaves the flow path filled with ...	UN-09	SSD-CSI-100; DWG-CSI-1000, -1100; SRS-SW: RA-202	D: DVP-601; PVR-CSI-001	Pass
SYS-603	The system shall provide automated cleaning modes: flow cell clean, sample line backflush ...	UN-09, UN-14	SSD-CSI-100; DWG-CSI-1000, -1100; SRS-SW: RA-203	D: DVP-601; PVR-CSI-001	Pass
SYS-604	The system shall be ready to run daily QC within 45 minutes of power-on, including laser ...	UN-04	DWG-CSI-0001; BOM-CSI-0001; SRS-SW: CM-302	T: DVP-601; PVR-CSI-001	Pass
SYS-611	Selecting a nozzle size in software shall load a verified default setup for sheath pressure ...	UN-09, UN-15	SSD-CSI-700; DWG-CSI-7000; SRS-SW: RA-401, RA-403, RA-405, AD-701	D: DVP-601; PVR-CSI-001	Pass
SYS-612	The system shall display a live image of the drop breakoff and of the side streams.	UN-09	SSD-CSI-700; DWG-CSI-7000; SRS-SW: RA-404, FW-209, AD-501	D: DVP-601; PVR-CSI-001	Pass

ID	Requirement (abbreviated)	Source	Design output	Verification	Result
SYS-613	The system shall hold the breakoff position automatically by adjusting drop drive amplitude.	UN-08	SSD-CSI-700; DWG-CSI-7000; SRS-SW: RA-501, RA-502, FW-209	T: DVP-601; PVR-CSI-001	Pass (after ANM-0014)
SYS-614	The system shall provide a test sort function that generates side streams for adjustment ...	UN-09	SSD-CSI-700; DWG-CSI-7000; SRS-SW: RA-406	D: DVP-601; PVR-CSI-001	Pass
SYS-621	The system shall run a daily bead-based performance check that measures detector performance ...	UN-08	SRS-SW: RA-801, RA-802, AD-601	T: DVP-601; PVR-CSI-001	Pass
SYS-622	The system shall not allow an IVD assay to run unless the most recent performance check for ...	UN-08, HA-08	SRS-SW: RA-701, RA-803, RA-804, AD-104	T: DVP-601; PVR-CSI-001	Pass (after ANM-0025)
SYS-623	The software configuration shall identify the installed lasers, mirrors and filters, and shall ...	UN-16, HA-08	SRS-SW: RA-805, FW-302, AD-101, AD-102, AD-103, AD-105, AD-106	T: DVP-601; PVR-CSI-001	Pass
SYS-624	Performance check results shall be stored and trended so users can see changes over time.	UN-08	SRS-SW: RA-806	D: DVP-601; PVR-CSI-001	Pass
SYS-631	The system shall allow users to define gates, sort populations and sort layouts for each ...	UN-02, UN-05	SRS-SW: RA-306, RA-602	D: DVP-601; PVR-CSI-001	Pass
SYS-632	The system shall calculate fluorescence compensation from single-stained controls.	UN-01, UN-03	SRS-SW: RA-305	T: DVP-601; PVR-CSI-001	Pass
SYS-633	The system shall show live sort counters for each sort population and the sort efficiency.	UN-02	SRS-SW: RA-702, RA-704	D: DVP-601; PVR-CSI-001	Pass
SYS-634	The user shall be able to pause, resume and stop a sort without losing the sort record.	UN-02	SRS-SW: RA-703, FW-207	D: DVP-601; PVR-CSI-001	Pass
SYS-635	The system shall produce a sort report giving the sort settings, counts and events for each ...	UN-10	SRS-SW: RA-307, RA-503, RA-706, RA-902, DM-101, DM-102	D: DVP-601; PVR-CSI-001	Pass
SYS-701	The software shall require a unique user login and support at least operator, supervisor and ...	UN-10	SRS-SW: AD-801, CM-101, CM-102, CM-103	T: DVP-701; PVR-CSI-001; SVR-SW-CSI-010	Pass
SYS-702	The software shall keep a secure, computer-generated, time-stamped audit trail of record ...	UN-10, 21 CFR 11	SRS-SW: FW-706, FW-808, DM-201, DM-202, DM-203, DM-204, DM-501, DM-502, DM-503	T: DVP-701; PVR-CSI-001; SVR-SW-CSI-010	Pass
SYS-703	The software shall support electronic signatures for result release and assay approval.	UN-10, 21 CFR 11	SRS-SW: RA-904, AD-103, AD-802, DM-301, DM-302, DM-303, DM-304	T: DVP-701; PVR-CSI-001; SVR-SW-CSI-010	Pass
SYS-704	Approved IVD assay templates shall be locked so that operators cannot change instrument ...	UN-01, HA-08	SRS-SW: RA-901, RA-902, AD-802	T: DVP-701; PVR-CSI-001; SVR-SW-CSI-010	Pass
SYS-705	The software shall export data in FCS 3.1 format and reports in PDF.	UN-10	SRS-SW: RA-307, DM-401, DM-404	T: DVP-701; PVR-CSI-001; SVR-SW-CSI-010	Pass

ID	Requirement (abbreviated)	Source	Design output	Verification	Result
SYS-706	The software shall detect and report loss of communication with the instrument without ...	UN-10	SRS-SW: FW-603, FW-804, FW-805, DM-103, DM-104, DM-105	T: DVP-701; PVR-CSI-001; SVR-SW-CSI-010	Pass
SYS-707	The workstation shall have a locked, validated operating system image, with only the services ...	Cybersecurity	SRS-SW: FW-104, FW-802, FW-803, CM-201, CM-202, CM-304	I: DVP-701; PVR-CSI-001; SVR-SW-CSI-010	Pass
SYS-708	The manufacturer shall maintain a software bill of materials and a process for assessing and ...	Cybersecurity	SRS-SW: FW-104, DM-204, CM-202, CM-203	I: DVP-701; PVR-CSI-001; SVR-SW-CSI-010	Pass
SYS-709	User data at rest shall be protected by access control, and data exported off the workstation ...	UN-10, Cybersecurity	SRS-SW: FW-705, DM-402, DM-403, DM-504, CM-103	T: DVP-701; PVR-CSI-001; SVR-SW-CSI-010	Pass
SYS-801	The sorter and fluidics supply module shall connect through a keyed umbilical carrying power ...	HA-14	SSD-CSI-100; DWG-CSI-1000, -1100	I: DVP-801; PVR-CSI-001	Pass
SYS-802	The sorter shall connect to the workstation by Ethernet on a dedicated, isolated network link.	UN-10	SSD-CSI-600; SCH-CSI-6000; SRS-SW: FW-801, CM-201	I: DVP-801; PVR-CSI-001	Pass
SYS-803	The system shall provide connections for the aerosol evacuation module and the temperature ...	UN-11, UN-06	SSD-CSI-100; DWG-CSI-1000, -1100; SSD-CSI-900; SRS-SW: RA-208	D: DVP-801; PVR-CSI-001	Pass
SYS-804	Collection tube holders shall share a universal top with evacuation ports compatible with the ...	UN-06	SSD-CSI-800	I: DVP-801; PVR-CSI-001	Pass
SYS-805	The system shall use sheath fluid of buffered saline or 1× PBS, and cleaning fluids of 10% ...	UN-09	SSD-CSI-100; DWG-CSI-1000, -1100	T: DVP-801; PVR-CSI-001	Pass
SYS-806	The system shall support IVD reagents and QC, alignment and drop-delay beads as specified in ...	UN-01, UN-08	DWG-CSI-0001; BOM-CSI-0001; SRS-SW: AD-602, AD-603	T: DVP-801; PVR-CSI-001	Pass
SYS-901	A trained operator shall be able to complete startup, daily QC, a four-way sort and shutdown ...	UN-09	DWG-CSI-0001; BOM-CSI-0001; SRS-SW: RA-101, RA-102, RA-105, RA-1001, CM-301	D: DVP-901; PVR-CSI-001	Pass
SYS-902	All critical tasks identified in the use-related risk analysis shall be completed without use ...	UN-06, IEC 62366-1	DWG-CSI-0001; BOM-CSI-0001; SRS-SW: CM-305	T: DVP-901; PVR-CSI-001	Pass (after ANM-0051)
SYS-903	Tube holders, nozzles and collection devices shall be marked with their size and keyed or ...	HA-08	SSD-CSI-100; SSD-CSI-300; DWG-CSI-3100; SSD-CSI-800	I: DVP-901; PVR-CSI-001	Pass
SYS-904	The software shall warn the user before any action that would discard unsaved data or stop an ...	UN-10	SRS-SW: RA-104	D: DVP-901; PVR-CSI-001	Pass
SYS-905	Fluid containers shall be labeled and color-coded by content, and their connectors keyed to ...	HA-14	SSD-CSI-100; DWG-CSI-1000, -1100	I: DVP-901; PVR-CSI-001	Pass
SYS-1001	The system shall have a design service life of at least 7 years with scheduled maintenance.	UN-12	DWG-CSI-0001; BOM-CSI-0001; SRS-SW: FW-704	A: DVP-1001; PVR-CSI-001	Pass

ID	Requirement (abbreviated)	Source	Design output	Verification	Result
SYS-100 2	Mean time between failures requiring a service visit shall be at least 2,000 operating hours.	UN-12	DWG-CSI-0001; BOM-CSI-0001; SRS-SW: CM-303	A, T: DVP-1001; PVR-CSI-001	Pass
SYS-100 3	Routine user maintenance (filters, sample lines, waste caps, nozzle cleaning) shall be ...	UN-12	DWG-CSI-0001; BOM-CSI-0001	D: DVP-1001; PVR-CSI-001	Pass
SYS-100 4	User maintenance intervals shall be no more frequent than weekly for filter purging and every ...	UN-12	SSD-CSI-100; DWG-CSI-1000, -1100; SRS-SW: AD-903	A: DVP-1001; PVR-CSI-001	Pass
SYS-100 5	Lasers, PMTs, detector arrays, PCBAs and the fluidics supply module shall be field-replaceable ...	UN-12	DWG-CSI-0001; BOM-CSI-0001; SRS-SW: FW-301, FW-704	D: DVP-1001; PVR-CSI-001	Pass
SYS-100 6	A trained service engineer shall be able to replace any field-replaceable unit and return the ...	UN-12	DWG-CSI-0001; BOM-CSI-0001; SRS-SW: AD-902, AD-904	D: DVP-1001; PVR-CSI-001	Pass
SYS-100 7	The system shall provide diagnostic data (pressures, laser status, voltages, error logs) to ...	UN-12, UN-10	SSD-CSI-600; SCH-CSI-6000; SRS-SW: FW-101, FW-310, FW-701, FW-702, FW-703, FW-705, AD-201, AD-901	D: DVP-1001; PVR-CSI-001	Pass
SYS-100 8	Lasers and detector arrays shall be upgradable in the field according to CFG-CSI-001.	UN-16	SSD-CSI-400; DWG-CSI-4000; SSD-CSI-500; DWG-CSI-5100, -5200; SRS-SW: AD-101, AD-107	D: DVP-1001; PVR-CSI-001	Pass
SYS-110 1	Product labels shall meet ISO 15223-1 and ISO 18113-3 and shall include the UDI in ...	UN-13	LBL-CSI-001; OG-CSI-001	I: DVP-1101; PVR-CSI-001	Pass
SYS-110 2	The IFU (Operator's Guide) shall include intended use, limitations, warnings, installation ...	UN-13, ISO 18113-3	LBL-CSI-001; OG-CSI-001	I: DVP-1101; PVR-CSI-001	Pass
SYS-110 3	Laser, high-voltage, biohazard and pinch-point labels shall be placed where the hazard exists ...	HA-01, HA-02, HA-06	LBL-CSI-001; OG-CSI-001	I: DVP-1101; PVR-CSI-001	Pass
SYS-110 4	The system shall be supplied with a site preparation guide and a service manual.	UN-07, UN-12	LBL-CSI-001; OG-CSI-001	I: DVP-1101; PVR-CSI-001	Pass
SYS-110 5	Labeling and IFU shall be available in the languages required by each market.	UN-13	LBL-CSI-001; OG-CSI-001; SRS-SW: CM-305	I: DVP-1101; PVR-CSI-001	Pass
SYS-120 1	The shipping configuration shall protect the sorter and the fluidics supply module so that ...	UN-07, UN-08	PKG-CSI-001; SPG-CSI-001	T: DVP-1201; PVR-CSI-001; ISTA report	Pass (after ANM-0040)
SYS-120 2	Shipping crates shall carry shock and tilt indicators.	UN-12	PKG-CSI-001; SPG-CSI-001	I: DVP-1201; PVR-CSI-001; ISTA report	Pass
SYS-120 3	Installation, including installation qualification, shall be completed by a service engineer ...	UN-07	PKG-CSI-001; SPG-CSI-001	D: DVP-1201; PVR-CSI-001; ISTA report	Pass

ID	Requirement (abbreviated)	Source	Design output	Verification	Result
SYS-120 4	The system shall pass through a standard 90 cm (36 in) laboratory door with covers or crate ...	UN-07	PKG-CSI-001; SPG-CSI-001	I: DVP-1201; PVR-CSI-001; ISTA report	Pass
SYS-130 1	The system shall be designed and manufactured under a quality management system that meets 21 ...	UN-13	DHF-CSI-000; SRS-SW: AD-905	I: DVP-1301; PVR-CSI-001	Pass
SYS-130 2	The system shall meet the requirements for US marketing as a Class II IVD instrument through ...	UN-13	DHF-CSI-000	A: DVP-1301; PVR-CSI-001	Pass
SYS-130 3	The system shall meet the general safety and performance requirements of EU IVDR 2017/746 ...	UN-13	DHF-CSI-000	A: DVP-1301; PVR-CSI-001	Pass
SYS-130 4	Risk management shall be carried out according to ISO 14971:2019 and all residual risks shall ...	UN-06, UN-13	DHF-CSI-000	A: DVP-1301; PVR-CSI-001	Pass
SYS-130 5	Software shall be developed according to IEC 62304 and IEC 82304-1.	UN-13	SRS-SW: CM-401	I: DVP-1301; PVR-CSI-001	Pass
SYS-130 6	The system shall comply with applicable restrictions on hazardous substances (EU RoHS) and ...	UN-13	DHF-CSI-000	I: DVP-1301; PVR-CSI-001	Pass

5. Matrix C: Software Requirements to Design and Tests

One row per software requirement in SRS-SW-CSI-010. Design gives the software item and the section of SAD-SW-CSI-011 that decomposes it into units; detailed design is in SDS-SW-CSI-012 (workstation) or SDS-FW-CSI-013 (firmware). Tests are at the level stated in the SRS-SW; all software requirements are also covered by the software system test (STS) part for their application. Results from SVR-SW-CSI-010.

ID	Requirement (abbreviated)	Parent	Item / design	Tests	Result
RA-101	The application shall provide a workspace with experiment browser, plots ...	SYS-901	SI-01; SAD §6.2	S: STS §A	Pass
RA-102	The application shall guide the operator through startup, daily QC, sorting and ...	SYS-901	SI-01; SAD §6.2	S: STS §A	Pass
RA-103	The application shall show fluid levels, system pressure, laser status, HV ...	SYS-493, SYS-522	SI-01; SAD §6.2	S: STS §A	Pass
RA-104	The application shall ask for confirmation before any action that discards ...	SYS-904	SI-01; SAD §6.2	S: STS §A	Pass
RA-105	The application shall show progress and remaining time for each automated ...	SYS-901	SI-01; SAD §6.2	S: STS §A	Pass
RA-201	The application shall provide an automated fluidics startup that pressurizes ...	SYS-601	SI-04; SAD §6.2	S: STS §A	Pass
RA-202	The application shall provide an automated fluidics shutdown that fills the ...	SYS-602	SI-04; SAD §6.2	S: STS §A	Pass
RA-203	The application shall provide cleaning modes: flow cell clean, sample line ...	SYS-603	SI-04; SAD §6.2	S: STS §A	Pass
RA-204	The application shall not start fluidics startup unless a nozzle is detected as ...	HA-09	SI-04, SI-10; SAD §6.2, §7.2	I: ITS §4, ITS §6, STS §A	Pass
RA-205	On Load, the application shall raise the loading port, pressurize the sample ...	SYS-472, SYS-494	SI-04, SI-10; SAD §6.2, §7.2	I: ITS §4, ITS §6, STS §A	Pass (after ANM-0019)
RA-206	The application shall offer flow rate settings 1 to 11 (about 10 to 80 µL/min) ...	SYS-453, SYS-495, SYS-4101	SI-04; SAD §6.2	S: STS §A	Pass
RA-207	The application shall warn when sheath falls below 20% or waste rises above ...	SYS-493	SI-04; SAD §6.2	S: STS §A	Pass
RA-208	The application shall detect the aerosol evacuation module, temperature control ...	SYS-803, SYS-551	SI-04; SAD §6.2	S: STS §A	Pass
RA-301	The application shall let the user set detector voltages, FSC gain, threshold ...	SYS-443	SI-02; SAD §6.2	S: STS §A	Pass
RA-302	The application shall display data on a linear scale from 2.6 to 262,144 or on ...	SYS-445	SI-02; SAD §6.2	U: UTS-CSI-010-02, STS §A	Pass

ID	Requirement (abbreviated)	Parent	Item / design	Tests	Result
RA-303	The application shall acquire and display at least 70,000 events per second ...	SYS-451, SYS-452	SI-02, SI-41; SAD §6.2, §6.5	S: STS §A	Pass
RA-304	The application shall report events lost to electronic abort or overflow and ...	SYS-451	SI-02; SAD §6.2	I: ITS §2, STS §A	Pass
RA-305	The application shall calculate a compensation matrix from single-stained ...	SYS-632	SI-02; SAD §6.2	U: UTS-CSI-010-02, STS §A	Pass
RA-306	The application shall let users define polygon, rectangle, interval and ...	SYS-631	SI-02; SAD §6.2	S: STS §A	Pass
RA-307	The application shall store acquired data with all instrument settings ...	SYS-635, SYS-705	SI-02, SI-31; SAD §6.2, §6.4	I: ITS §2, ITS §16, STS §A	Pass
RA-401	The application shall load the controlled default sort setup for the selected ...	SYS-611	SI-03; SAD §6.2	S: STS §A	Pass
RA-402	Changing the sort setup shall require confirmation of the installed nozzle size.	HA-08	SI-03; SAD §6.2	S: STS §A	Pass
RA-403	The application shall restore the last setup at startup with stream, breakoff ...	SYS-611, HA-02	SI-03; SAD §6.2	I: ITS §3, STS §A	Pass
RA-404	The application shall display the breakoff and side-stream images at least 10 ...	SYS-612	SI-03, SI-13; SAD §6.2, §7.2	I: ITS §3, ITS §9, STS §A	Pass
RA-405	The application shall let the user set drop drive frequency (1.0 to 102.0 kHz) ...	SYS-463, SYS-611	SI-03; SAD §6.2	S: STS §A	Pass
RA-406	The application shall let the user turn plate voltage on and off, run a test ...	SYS-614	SI-03; SAD §6.2	S: STS §A	Pass
RA-407	The application shall determine drop delay automatically with drop-delay beads ...	SYS-465	SI-03, SI-13; SAD §6.2, §7.2	S: STS §A	Pass
RA-501	When breakoff control is On, the application shall adjust drop drive amplitude ...	SYS-613	SI-03, SI-13; SAD §6.2, §7.2	I: ITS §3, ITS §9, STS §A	Pass (after ANM-0014)
RA-502	If Gap differs from its target by more than 1 pixel, the application shall ...	SYS-613	SI-03; SAD §6.2	I: ITS §3, STS §A	Pass
RA-503	The application shall record each sort pause and resume commanded by firmware ...	SYS-635	SI-03; SAD §6.2	I: ITS §3, STS §A	Pass
RA-504	After a clog, the application shall show the clog recovery procedure, including ...	SYS-554, HA-04	SI-01; SAD §6.2	S: STS §A	Pass (after ANM-0051)
RA-505	The application shall not allow the stream or plate voltage to restart after a ...	HA-04	SI-03, SI-10; SAD §6.2, §7.2	I: ITS §3, ITS §6, STS §A	Pass
RA-601	The application shall provide sort layouts for two-way 12 × 75 mm and 15 mL ...	SYS-481, SYS-482, SYS-485	SI-03; SAD §6.2	S: STS §A	Pass

ID	Requirement (abbreviated)	Parent	Item / design	Tests	Result
RA-602	The application shall let the user assign gated populations to each sort ...	SYS-631	SI-03; SAD §6.2	S: STS §A	Pass
RA-603	The application shall not allow a four-way layout with a two-way device, or a ...	HA-08	SI-03; SAD §6.2	U: UTS-CSI-010-03, STS §A	Pass
RA-604	The application shall provide the read-only precision modes in Table 1 and any ...	SYS-464	SI-03; SAD §6.2	S: STS §A	Pass
RA-605	Yield and phase masks shall be mutually exclusive: setting one above 0 shall ...	SYS-464	SI-03; SAD §6.2	U: UTS-CSI-010-03, STS §A	Pass
RA-701	The application shall not start a sort unless the stream is on, breakoff ...	SYS-622, HA-08	SI-03, SI-05; SAD §6.2	S: STS §A	Pass
RA-702	The application shall display, for each sort location, sorted count, sort rate ...	SYS-633	SI-01, SI-03; SAD §6.2	S: STS §A	Pass
RA-703	The user shall be able to pause, resume and stop a sort; stopping shall close ...	SYS-634	SI-03; SAD §6.2	I: ITS §3, STS §A	Pass
RA-704	The application shall stop the sort automatically when all target counts are ...	SYS-633	SI-03; SAD §6.2	S: STS §A	Pass
RA-705	In index sorting, the application shall store the parameters of each deposited ...	SYS-484	SI-03, SI-31; SAD §6.2, §6.4	I: ITS §3, ITS §16, STS §A	Pass
RA-706	At the end of each sort the application shall create a sort report with ...	SYS-635	SI-03, SI-31; SAD §6.2, §6.4	S: STS §A	Pass
RA-801	The application shall run the bead-based performance check against the baseline ...	SYS-621	SI-05; SAD §6.2	S: STS §A	Pass
RA-802	The performance check shall set inter-laser delay and area scaling ...	SYS-444, SYS-621	SI-05, SI-11; SAD §6.2, §7.2	I: ITS §5, ITS §7, STS §A	Pass
RA-803	The application shall not allow acquisition or sorting with an IVD assay ...	SYS-622, HA-08	SI-05; SAD §6.2	S: STS §A	Pass (after ANM-0025)
RA-804	A supervisor shall be able to record a documented QC lockout override for ...	SYS-622	SI-05, SI-40; SAD §6.2, §6.5	S: STS §A	Pass
RA-805	The application shall warn the user before acquisition if the active lasers do ...	SYS-623, HA-08	SI-05, SI-10; SAD §6.2, §7.2	I: ITS §5, ITS §6, STS §A	Pass
RA-806	The application shall display trends of detector voltage, robust CV and laser ...	SYS-624	SI-05; SAD §6.2	S: STS §A	Pass
RA-901	When running an approved IVD assay template, the application shall lock ...	SYS-704, HA-08	SI-05, SI-02; SAD §6.2	S: STS §A	Pass

ID	Requirement (abbreviated)	Parent	Item / design	Tests	Result
RA-902	Results from an IVD assay template shall show the template name and version and ...	SYS-704, SYS-635	SI-05; SAD §6.2	S: STS §A	Pass
RA-903	The application shall mark any result acquired while a QC, configuration or ...	HA-08	SI-02, SI-05; SAD §6.2	S: STS §A	Pass
RA-904	The application shall send completed IVD results to the Data Management ...	SYS-703	SI-05, SI-31; SAD §6.2, §6.4	I: ITS §5, ITS §16, STS §A	Pass
RA-1001	The application shall classify alarms as high, medium and low, and show each ...	SYS-901	SI-01; SAD §6.2	S: STS §A	Pass
RA-1002	High alarms shall include clog, emergency stop, loss of instrument ...	HA-04, HA-06, HA-15	SI-01, SI-04; SAD §6.2	S: STS §A	Pass (after ANM-0023, ANM-0048)
RA-1003	The application shall show the HV-on indicator whenever firmware reports plate ...	SYS-522, HA-02	SI-01; SAD §6.2	I: ITS §1, STS §A	Pass
FW-101	On power-up, SI-10 shall run a power-on self test of memory, clocks, board bus ...	SYS-1007	SI-10; SAD §7.2	I: ITS §6, STS §B	Pass
FW-102	Each FPGA shall load its configuration only after verifying its CRC; a failed ...	HA-08	SI-11, SI-12; SAD §7.2	U: UTS-CSI-010-11, UTS-CSI-010-12, STS §B	Pass
FW-103	Firmware shall start in the safe state (Section 5.6) and leave it only on ...	HA-02	SI-10, SI-15; SAD §7.2	I: ITS §6, ITS §11, STS §B	Pass (after ANM-0007)
FW-104	Firmware and FPGA images shall be accepted for update only if their digital ...	SYS-707, SYS-708	SI-10; SAD §7.2	I: ITS §6, STS §B	Pass
FW-201	SI-11 shall compute height, area and width for every enabled parameter of every ...	SYS-443, SYS-444	SI-11; SAD §7.2	U: UTS-CSI-010-11, STS §B	Pass
FW-202	SI-11 shall digitize signals on an 18-bit linear scale and process at least ...	SYS-445, SYS-451, SYS-452	SI-11; SAD §7.2	U: UTS-CSI-010-11, STS §B	Pass
FW-203	SI-11 shall resolve the position of each event within its drop to 1/32 of a ...	SYS-464	SI-11; SAD §7.2	U: UTS-CSI-010-11, STS §B	Pass
FW-204	SI-11 shall complete the sort decision for each event within 20 μ s of the end ...	SYS-461	SI-11; SAD §7.2	U: UTS-CSI-010-11, STS §B	Pass (after ANM-0003)
FW-205	During sorting, if Gap differs from its target by more than 2 pixels or Drop 1 ...	SYS-542, HA-07	SI-11, SI-13; SAD §7.2	I: ITS §7, ITS §9, STS §B	Pass
FW-206	Single-cell modes shall sort at most one target event per drop and count every ...	SYS-483	SI-11; SAD §7.2	U: UTS-CSI-010-11, STS §B	Pass
FW-207	Pause shall stop charging within 100 ms while acquisition continues; resume ...	SYS-634	SI-11; SAD §7.2	I: ITS §7, STS §B	Pass

ID	Requirement (abbreviated)	Parent	Item / design	Tests	Result
FW-208	SI-10 shall move the plate stage to each well in layout order and enable ...	SYS-482, SYS-483	SI-10; SAD §7.2	I: ITS §6, STS §B	Pass
FW-209	SI-13 shall capture strobed breakoff images at least 10 times per second and ...	SYS-612, SYS-613	SI-13; SAD §7.2	U: UTS-CSI-010-13, STS §B	Pass
FW-210	SI-13 shall detect a clog (loss or deflection of the stream, or Gap above the ...	SYS-541, HA-04	SI-13; SAD §7.2	U: UTS-CSI-010-13, STS §B	Pass
FW-301	Each instrument board (ADC, PMT high voltage, laser control, deflection HV ...	SYS-1005	SI-12; SAD §7.2	I: ITS §8, STS §B	Pass
FW-302	At startup, SI-10 shall read every board identity, compare the inventory with ...	SYS-623, HA-08	SI-10, SI-12; SAD §7.2	I: ITS §6, ITS §8, STS §B	Pass
FW-303	SI-12 shall configure the ADC boards (sample rate, gain, offset) from the board ...	SYS-443	SI-12; SAD §7.2	U: UTS-CSI-010-12, STS §B	Pass
FW-304	SI-12 shall set each PMT high voltage from 0 to 1,000 V as commanded, read it ...	SYS-441, HA-08	SI-12; SAD §7.2	U: UTS-CSI-010-12, STS §B	Pass (after ANM-0034)
FW-305	SI-12 shall switch each laser on and off, set laser power where the laser ...	SYS-433, SYS-435	SI-12; SAD §7.2	U: UTS-CSI-010-12, STS §B	Pass
FW-306	SI-12 shall enforce that lasers sharing a beam spot (488/445 and 405/375) are ...	SYS-435	SI-12; SAD §7.2	U: UTS-CSI-010-12, STS §B	Pass
FW-307	SI-12 shall set deflection plate voltage and stream charge levels as commanded ...	SYS-523, HA-02	SI-12; SAD §7.2	U: UTS-CSI-010-12, STS §B	Pass; open: ANM-0066
FW-308	SI-12 shall monitor supply rails (within $\pm 5\%$), board temperatures and fan ...	HA-12	SI-12; SAD §7.2	U: UTS-CSI-010-12, STS §B	Pass
FW-309	SI-12 shall drive the loading port, aspirator drawer, drop-delay filter and ...	SYS-562	SI-12; SAD §7.2	U: UTS-CSI-010-12, STS §B	Pass
FW-310	SI-12 shall expose board status and counters through the register map defined ...	SYS-1007	SI-12; SAD §7.2	I: ITS §8, STS §B	Pass
FW-401	SI-14 shall hold sheath pressure within $\pm 1\%$ of the setpoint across 5 to 75 PSI ...	SYS-491	SI-14; SAD §7.2	U: UTS-CSI-010-14, STS §B	Pass
FW-402	SI-14 shall report sheath, waste, ethanol and cleaning fluid levels and supply ...	SYS-493	SI-14, SI-10; SAD §7.2	I: ITS §10, ITS §6, STS §B	Pass
FW-403	On main power off, SI-14 shall vent system pressure within 10 seconds.	SYS-524	SI-14; SAD §7.2	I: ITS §10, STS §B	Pass
FW-501	The loading port shall move only while the emergency stop is not active, and ...	SYS-533	SI-10; SAD §7.2	I: ITS §6, STS §B	Pass

ID	Requirement (abbreviated)	Parent	Item / design	Tests	Result
FW-502	SI-10 shall monitor door interlock, laser shutter and emergency stop status and ...	SYS-512, SYS-521	SI-10; SAD §7.2	I: ITS §6, STS §B	Pass
FW-601	On emergency stop, SI-10 shall unload the sample tube, stop the plate stage ...	SYS-532, HA-02, HA-06	SI-10; SAD §7.2	I: ITS §6, STS §B	Pass
FW-602	On a detected clog, SI-10 shall, without waiting for the workstation, stop the ...	SYS-543, HA-04	SI-10; SAD §7.2	I: ITS §6, STS §B	Pass (after ANM-0011)
FW-603	If communication with the workstation is lost for more than 2 seconds during ...	SYS-706, HA-02	SI-10; SAD §7.2	I: ITS §6, STS §B	Pass
FW-604	SI-15 shall monitor heartbeats from SI-10 and SI-11; if either is missing for ...	HA-02	SI-15; SAD §7.2	U: UTS-CSI-010-15, STS §B	Pass (after ANM-0007)
FW-605	SI-10 and SI-15 shall each run a hardware watchdog; a watchdog timeout shall ...	HA-02	SI-10, SI-15; SAD §7.2	U: UTS-CSI-010-10, UTS-CSI-010-15, STS §B	Pass
FW-606	A board fault reported under FW-304, FW-307 or FW-308, an FPGA configuration ...	HA-02, HA-08, HA-12	SI-10, SI-12; SAD §7.2	I: ITS §6, ITS §8, STS §B	Pass
FW-607	The safe state shall be latched: firmware shall leave it only after the cause ...	HA-02, HA-04	SI-10; SAD §7.2	I: ITS §6, STS §B	Pass
FW-608	SI-10 shall report HV status at least 5 times per second and report every entry ...	SYS-522	SI-10; SAD §7.2	I: ITS §6, STS §B	Pass
FW-701	SI-10 shall keep an internal log of events, warnings, faults, safe-state ...	SYS-1007	SI-10; SAD §7.2	I: ITS §6, STS §B	Pass
FW-702	Each log entry shall contain a UTC time stamp with millisecond resolution ...	SYS-1007	SI-10; SAD §7.2	U: UTS-CSI-010-10, STS §B	Pass
FW-703	The log shall be kept in non-volatile memory for at least 100,000 entries or 90 ...	SYS-1007	SI-10; SAD §7.2	I: ITS §6, STS §B	Pass (after ANM-0035)
FW-704	SI-10 shall keep usage counters for laser on-hours per laser, PMT on-hours ...	SYS-1001, SYS-1005	SI-10; SAD §7.2	U: UTS-CSI-010-10, STS §B	Pass
FW-705	The log shall contain no patient or specimen data, and shall be readable and ...	SYS-1007, SYS-709	SI-10; SAD §7.2	I: ITS §6, STS §B	Pass
FW-706	Log entries shall not be editable or deletable by any user; the log export ...	SYS-702	SI-10; SAD §7.2	U: UTS-CSI-010-10, STS §B	Pass
FW-801	The instrument shall communicate with the workstation over a dedicated Ethernet ...	SYS-802	SI-10, SI-41; SAD §7.2, §6.5	I: ITS §6, ITS §19, STS §B	Pass
FW-802	The instrument shall accept connections only from the paired workstation ...	SYS-707	SI-10, SI-41; SAD §7.2, §6.5	I: ITS §6, ITS §19, STS §B	Pass (after ANM-0030)

ID	Requirement (abbreviated)	Parent	Item / design	Tests	Result
FW-803	Every message shall carry a sequence number and a CRC-32; messages with a bad ...	SYS-707, HA-08	SI-10; SAD §7.2	U: UTS-CSI-010-10, STS §B	Pass (after ANM-0030)
FW-804	The instrument shall acknowledge every command within 100 ms with accepted ...	SYS-706	SI-10; SAD §7.2	U: UTS-CSI-010-10, STS §B	Pass
FW-805	Instrument and workstation shall exchange heartbeats every 250 ms; loss for ...	SYS-706	SI-10, SI-41; SAD §7.2, §6.5	I: ITS §6, ITS §19, STS §B	Pass (after ANM-0023)
FW-806	The event data channel shall carry at least 110,000 events per second at 8 ...	SYS-451, SYS-452	SI-10, SI-11, SI-41; SAD §7.2, §6.5	I: ITS §6, ITS §7, ITS §19, STS §B	Pass
FW-807	Workstation and instrument shall check protocol version compatibility at ...	HA-08	SI-10, SI-41; SAD §7.2, §6.5	I: ITS §6, ITS §19, STS §B	Pass
FW-808	The instrument clock shall be synchronized to the workstation at connection and ...	SYS-702	SI-10, SI-41; SAD §7.2, §6.5	I: ITS §6, ITS §19, STS §B	Pass
AD-101	The application shall maintain a controlled hardware configuration record for ...	SYS-623, SYS-1008	SI-20; SAD §6.3	S: STS §C	Pass
AD-102	The application shall compare the configuration record with the board inventory ...	SYS-623, HA-08	SI-20, SI-10; SAD §6.3, §7.2	I: ITS §12, ITS §6, STS §C	Pass
AD-103	Each change to the configuration record shall require the administrator or ...	SYS-623, SYS-703	SI-20, SI-32; SAD §6.3, §6.4	S: STS §C	Pass
AD-104	After a configuration change affecting lasers, optics or detectors, the ...	SYS-622, HA-08	SI-20, SI-05; SAD §6.3, §6.2	S: STS §C	Pass
AD-105	The application shall enforce the configuration rules in CFG-CSI-001 ...	SYS-623	SI-20; SAD §6.3	U: UTS-CSI-010-20, STS §C	Pass (after ANM-0026)
AD-106	The application shall generate the optical configuration (parameter names ...	SYS-623	SI-20; SAD §6.3	I: ITS §12, STS §C	Pass
AD-107	The application shall record field upgrades of lasers and detector arrays as ...	SYS-1008	SI-20; SAD §6.3	S: STS §C	Pass
AD-201	The application shall display, for each laser, measured power, current ...	SYS-433, SYS-1007	SI-21; SAD §6.3	S: STS §C	Pass
AD-202	The application shall provide a laser power calibration that records power ...	SYS-433	SI-21; SAD §6.3	S: STS §C	Pass (after ANM-0016)
AD-203	The application shall flag any laser whose power at the head is below the ...	SYS-433, HA-08	SI-21, SI-05; SAD §6.3, §6.2	S: STS §C	Pass
AD-204	The application shall provide a warm-up verification that records laser power ...	SYS-437	SI-21; SAD §6.3	S: STS §C	Pass

ID	Requirement (abbreviated)	Parent	Item / design	Tests	Result
AD-205	The application shall calibrate inter-laser delay and area scaling for each ...	SYS-444	SI-21; SAD §6.3	S: STS §C	Pass
AD-301	The application shall provide an alignment verification tool that runs ...	SYS-436	SI-21; SAD §6.3	S: STS §C	Pass
AD-302	The application shall calibrate each PMT's voltage-to-gain curve with ...	SYS-441	SI-21; SAD §6.3	S: STS §C	Pass
AD-303	The application shall verify detector linearity with multi-peak beads and ...	SYS-411	SI-21; SAD §6.3	S: STS §C	Pass
AD-304	The application shall provide a resolution check with chicken erythrocyte ...	SYS-401, SYS-402, SYS-411	SI-21; SAD §6.3	S: STS §C	Pass
AD-305	The application shall record the installed FSC ND filter and apply the matching ...	SYS-423	SI-21, SI-20; SAD §6.3	S: STS §C	Pass
AD-401	The application shall calibrate each pressure sensor at two points against a ...	SYS-491	SI-21, SI-14; SAD §6.3, §7.2	I: ITS §13, ITS §10, STS §C	Pass
AD-402	The application shall calibrate sample flow rate by recording the sample volume ...	SYS-453	SI-21; SAD §6.3	S: STS §C	Pass
AD-403	The application shall calibrate each level sensor at empty and full.	SYS-493	SI-21, SI-14; SAD §6.3, §7.2	I: ITS §13, ITS §10, STS §C	Pass
AD-404	The application shall run a pressure decay leak test of the sheath and sample ...	SYS-491	SI-21; SAD §6.3	S: STS §C	Pass
AD-405	The application shall calibrate the sample chamber temperature sensor against a ...	SYS-4101	SI-21; SAD §6.3	S: STS §C	Pass
AD-501	The application shall calibrate breakoff and side-stream camera scale and focus ...	SYS-612	SI-21, SI-13; SAD §6.3, §7.2	I: ITS §13, ITS §9, STS §C	Pass
AD-502	The application shall verify drop drive frequency and plate voltage against ...	SYS-463, SYS-523	SI-21, SI-12; SAD §6.3, §7.2	I: ITS §13, ITS §8, STS §C	Pass
AD-503	The application shall measure stream position in the waste aspirator for each ...	SYS-468	SI-21; SAD §6.3	S: STS §C	Pass
AD-504	The application shall calibrate the plate stage home position and the first and ...	SYS-483	SI-21, SI-12; SAD §6.3, §7.2	S: STS §C	Pass (after ANM-0037)
AD-505	The application shall verify collection temperature control against a reference ...	SYS-4102	SI-21; SAD §6.3	S: STS §C	Pass
AD-601	The application shall create a performance check baseline for each optical ...	SYS-621	SI-22, SI-21; SAD §6.3	S: STS §C	Pass
AD-602	The application shall import bead lot files with target values and limits, and ...	SYS-806, HA-08	SI-22; SAD §6.3	U: UTS-CSI-010-22, STS §C	Pass

ID	Requirement (abbreviated)	Parent	Item / design	Tests	Result
AD-603	The application shall manage drop-delay bead and alignment bead lots with ...	SYS-806	SI-22; SAD §6.3	S: STS §C	Pass
AD-701	The application shall hold the default sort setups for each nozzle as ...	SYS-611	SI-22, SI-32; SAD §6.3, §6.4	S: STS §C	Pass
AD-702	The application shall let an administrator define custom collection devices by ...	SYS-485	SI-22; SAD §6.3	S: STS §C	Pass
AD-703	The application shall let an administrator create, name and delete additional ...	SYS-464	SI-22; SAD §6.3	S: STS §C	Pass
AD-801	The application shall let an administrator create, disable and assign roles to ...	SYS-701	SI-22, SI-40; SAD §6.3, §6.5	S: STS §C	Pass
AD-802	The application shall let a supervisor approve an IVD assay template by ...	SYS-704, SYS-703	SI-22, SI-32; SAD §6.3, §6.4	S: STS §C	Pass
AD-901	The application shall display firmware diagnostics (pressures, laser status ...	SYS-1007	SI-21; SAD §6.3	S: STS §C	Pass
AD-902	Each calibration shall be recorded with as-found and as-left values, pass or ...	SYS-1006	SI-21, SI-31; SAD §6.3, §6.4	S: STS §C	Pass
AD-903	The application shall warn when a calibration is due within 30 days and block ...	SYS-1004, HA-08	SI-21, SI-05; SAD §6.3, §6.2	S: STS §C	Pass
AD-904	The application shall produce a calibration certificate report for each ...	SYS-1006	SI-21; SAD §6.3	S: STS §C	Pass
AD-905	Calibration tools used in manufacturing shall produce records in the format ...	SYS-1301	SI-21; SAD §6.3	S: STS §C	Pass
DM-101	The application shall let users browse and search experiments, tubes, sort ...	SYS-635	SI-30; SAD §6.4	S: STS §D	Pass
DM-102	The application shall display stored data, plots and statistics without an ...	SYS-635	SI-30; SAD §6.4	S: STS §D	Pass
DM-103	SI-31 shall use transactions so each record is written completely or not at ...	SYS-706	SI-31; SAD §6.4	U: UTS-CSI-010-31, STS §D	Pass
DM-104	On workstation power loss or crash, data recorded up to the last completed ...	SYS-706	SI-31; SAD §6.4	S: STS §D	Pass
DM-105	The application shall warn the administrator when database storage exceeds 80% ...	SYS-706	SI-31, SI-30; SAD §6.4	S: STS §D	Pass
DM-201	SI-32 shall record every creation, modification and deletion of experiments ...	SYS-702	SI-32; SAD §6.4	I: ITS §17, STS §D	Pass (after ANM-0028)

ID	Requirement (abbreviated)	Parent	Item / design	Tests	Result
DM-202	Audit trail entries shall be generated by the software within 1 second of the ...	SYS-702	SI-32; SAD §6.4	U: UTS-CSI-010-32, STS §D	Pass
DM-203	Supervisors shall be able to view, filter and export the audit trail.	SYS-702	SI-30; SAD §6.4	S: STS §D	Pass
DM-204	Security events (logins, failed logins, account changes, update attempts) shall ...	SYS-702, SYS-708	SI-32, SI-40; SAD §6.4, §6.5	S: STS §D	Pass
DM-301	IVD results shall have status Draft, Reviewed or Released; only released ...	SYS-703	SI-30; SAD §6.4	S: STS §D	Pass
DM-302	Electronic signatures shall require re-entry of the password and record the ...	SYS-703	SI-32; SAD §6.4	S: STS §D	Pass
DM-303	Signatures shall be linked to their records so that they cannot be copied or ...	SYS-703	SI-32; SAD §6.4	U: UTS-CSI-010-32, STS §D	Pass
DM-304	Changing a released record shall create a new version with reason and ...	SYS-703	SI-30, SI-32; SAD §6.4	S: STS §D	Pass
DM-401	The application shall export data in FCS 3.1 format with all keywords needed to ...	SYS-705	SI-30; SAD §6.4	U: UTS-CSI-010-30, STS §D	Pass (after ANM-0045)
DM-402	Exported files shall include a checksum so that later changes can be detected.	SYS-709	SI-30; SAD §6.4	U: UTS-CSI-010-30, STS §D	Pass
DM-403	The application shall offer an export option that removes patient identifiers.	SYS-709	SI-30; SAD §6.4	S: STS §D	Pass
DM-404	The application shall export released IVD results to a laboratory information ...	SYS-705	SI-30; SAD §6.4	S: STS §D	Pass (after ANM-0058)
DM-501	The application shall provide scheduled and manual backup of the database and ...	SYS-702	SI-30; SAD §6.4	S: STS §D	Pass
DM-502	The application shall archive records to external storage with integrity ...	SYS-702	SI-30; SAD §6.4	S: STS §D	Pass
DM-503	Records shall be retained for the periods in Section 9; deletion shall be ...	SYS-702	SI-30, SI-32; SAD §6.4	S: STS §D	Pass
DM-504	User data at rest shall be encrypted and accessible only through the ...	SYS-709	SI-31; SAD §6.4	I: ITS §16, STS §D	Pass
CM-101	All applications shall require login with a unique user ID and password through ...	SYS-701	SI-40; SAD §6.5	S: STS §E	Pass
CM-102	SI-40 shall provide operator, supervisor, administrator and service roles with ...	SYS-701	SI-40; SAD §6.5	S: STS §E	Pass

ID	Requirement (abbreviated)	Parent	Item / design	Tests	Result
CM-103	Passwords shall have configurable length, complexity and expiry; accounts shall ...	SYS-701, SYS-709	SI-40; SAD §6.5	U: UTS-CSI-010-40, STS §E	Pass (after ANM-0042)
CM-201	The workstation image shall enable only the services and ports needed for ...	SYS-707, SYS-802	SI-41; SAD §6.5	S: STS §E	Pass
CM-202	Application updates shall be accepted only if their digital signature is valid.	SYS-707, SYS-708	SI-30; SAD §6.4	I: ITS §15, STS §D	Pass
CM-203	A software bill of materials covering applications and firmware shall be ...	SYS-708	SI-10, SI-30; SAD §7.2, §6.4	S: STS §B	Pass
CM-301	Applications shall update plots at least twice per second at the maximum event ...	SYS-901	SI-01; SAD §6.2	S: STS §A	Pass
CM-302	Each application shall start and, where needed, connect to the instrument ...	SYS-604	SI-01, SI-41; SAD §6.2, §6.5	S: STS §A	Pass
CM-303	The software system shall run 24 hours of continuous acquisition at the maximum ...	SYS-1002	SI-02, SI-10; SAD §6.2, §7.2	S: STS §A	Pass (after ANM-0044)
CM-304	Applications shall run on the validated workstation configuration in the ...	SYS-707	SI-30; SAD §6.4	S: STS §D	Pass
CM-305	User interface text shall be stored in resource files to support translation ...	SYS-1105, SYS-902	SI-01; SAD §6.2	I: ITS §1, STS §A	Pass; open: ANM-0053
CM-401	All software items shall be developed, verified, released and maintained under ...	SYS-1305	SI-01 to SI-41 (all items); SAD §6.2, §6.5	S: STS §A	Pass

6. Matrix D: Risk Controls to Verification

Each hazard in HA-CSI-001 that is controlled by design traces to its system and software requirements and to the evidence that the control is implemented and effective (VVP-CSI-001 Section 8). Residual risk evaluation is in RMR-CSI-001.

Hazard	System requirements	Software requirements	Implementation evidence	Effectiveness evidence	Result
HA-01 Laser exposure	SYS-511 to 513	FW-502	DVP-CSI-502; SVR-CSI-001 (IEC 60825-1)	Accessible emission measurement with doors open and closed	Pass (after ANM-0032)
HA-02 High voltage	SYS-522, 523, 532	FW-307, FW-601, FW-604 to 608, RA-1003	DVP-CSI-503; STS-CSI-010 §B	Fault injection: SI-10 hang, heartbeat loss, readback fault; HV removed in every trial	Pass (after ANM-0007, ANM-0023, ANM-0048, ANM-0066)
HA-04 Aerosol after clog	SYS-541, 543, 551 to 554	FW-210, FW-602, RA-504, RA-505	DVP-CSI-503, DVP-CSI-504	Aerosol challenge with simulated clog; no escape	Pass (after ANM-0011, ANM-0051)
HA-05 Access to sort chamber	SYS-521	FW-502	DVP-CSI-503	Door sequence test, 59 trials	Pass
HA-06 Moving parts	SYS-531 to 533, 562	FW-501, FW-601, FW-309	DVP-CSI-503, DVP-CSI-505	Pinch force measurement; e-stop timing	Pass
HA-07 Breakoff drift, low purity	SYS-542	FW-205, RA-501, RA-502	DVP-CSI-503; ITS-CSI-010	Induced breakoff drift; charging stops within 100 ms; purity maintained	Pass (after ANM-0003)
HA-08 Wrong result (configuration, QC, calibration)	SYS-622, 623, 704, 903	FW-302, AD-102 to 105, AD-203, AD-903, RA-803, RA-805, RA-901	DVP-CSI-601, DVP-CSI-701; STS-CSI-010 §A, §C	Seeded errors (wrong filter, missing QC, overdue calibration, mismatched laser) are blocked	Pass (after ANM-0012, ANM-0016, ANM-0025, ANM-0026, ANM-0030, ANM-0040)
HA-09 Overpressure	SYS-524, 561	FW-401, FW-403	DVP-CSI-505	Relief valve test; depressurization within 10 s	Pass
HA-10 Chemical and decontamination	SYS-553	—	DVP-CSI-504	Material compatibility after 500 decontamination cycles	Pass
HA-11 Tip-over	SYS-563	—	DVP-CSI-505	10° tilt test	Pass
HA-12 Electrical shock and fire	SYS-314, 315, 501, 502	FW-308, FW-606	DVP-CSI-501; SVR-CSI-001 (IEC 61010)	Accredited laboratory results	Pass
HA-13 Electromagnetic interference	SYS-503	—	DVP-CSI-501; SVR-CSI-001 (IEC 61326)	Immunity tests with performance monitoring	Pass (after ANM-0021)
HA-14 Fluid misconnection	SYS-801, 905	—	DVP-CSI-801, DVP-CSI-901	Misconnection attempts not possible	Pass
HA-15 Fluid run-out during sort	SYS-493	RA-207, FW-402	DVP-CSI-408	Low sheath and full waste scenarios blocked before sort	Pass

7. Backward Trace: Tests to Requirements

Every test must be justified by a requirement. These tables trace each verification protocol, software test specification and validation study back to what it verifies or validates.

7.1 System verification protocols to system requirements

Protocol	Title	System requirements verified	Report
DVP-CSI-301	Physical, electrical and environmental	301, 302, 303, 304, 305, 306, 311, 312, 313, 314, 315, 316, 321, 322, 323, 324, 325, 326, 331 (19)	PVR-CSI-001
DVP-CSI-401	Fluorescence resolution and linearity	401, 402, 411 (3)	PVR-CSI-001
DVP-CSI-402	Scatter sensitivity and detection	421, 422, 423, 441, 442, 443, 444, 445 (8)	PVR-CSI-001
DVP-CSI-403	Excitation (lasers, beam geometry, warm-up)	431, 432, 433, 434, 435, 436, 437, 438 (8)	PVR-CSI-001
DVP-CSI-404	Acquisition rate and sample flow	451, 452, 453 (3)	PVR-CSI-001
DVP-CSI-405	Sort performance (purity, yield, viability, nozzles)	461, 462, 463, 464, 465, 466, 467, 468 (8)	PVR-CSI-001
DVP-CSI-406	Carryover	471, 472 (2)	PVR-CSI-001
DVP-CSI-407	Sort collection and plate deposition	481, 482, 483, 484, 485 (5)	PVR-CSI-001
DVP-CSI-408	Fluidics and temperature control	491, 492, 493, 494, 495, 4101, 4102 (7)	PVR-CSI-001
DVP-CSI-501	Electrical safety and EMC	501, 502, 503 (3)	SVR-CSI-001
DVP-CSI-502	Laser safety	511, 512, 513 (3)	SVR-CSI-001
DVP-CSI-503	Interlocks, HV, e-stop, clog and breakoff response	521, 522, 523, 524, 531, 532, 533, 541, 542, 543 (10)	PVR-CSI-001
DVP-CSI-504	Biohazard containment and decontamination	551, 552, 553, 554 (4)	PVR-CSI-001
DVP-CSI-505	Mechanical and pressure safety	561, 562, 563 (3)	PVR-CSI-001
DVP-CSI-601	Functional requirements	601, 602, 603, 604, 611, 612, 613, 614, 621, 622, 623, 624, 631, 632, 633, 634, 635 (17)	PVR-CSI-001
DVP-CSI-701	Software, data integrity and cybersecurity	701, 702, 703, 704, 705, 706, 707, 708, 709 (9)	PVR-CSI-001; SVR-SW-CSI-010
DVP-CSI-801	Interfaces	801, 802, 803, 804, 805, 806 (6)	PVR-CSI-001
DVP-CSI-901	Usability requirements (design verification)	901, 902, 903, 904, 905 (5)	PVR-CSI-001
DVP-CSI-1001	Reliability, maintainability and serviceability	1001, 1002, 1003, 1004, 1005, 1006, 1007, 1008 (8)	PVR-CSI-001
DVP-CSI-1101	Labeling and documentation	1101, 1102, 1103, 1104, 1105 (5)	PVR-CSI-001
DVP-CSI-1201	Packaging, transport and installation	1201, 1202, 1203, 1204 (4)	PVR-CSI-001; ISTA report
DVP-CSI-1301	Regulatory and standards compliance	1301, 1302, 1303, 1304, 1305, 1306 (6)	PVR-CSI-001

7.2 Software test specifications to software requirements

Test specification	Scope	Software requirements
UTS-CSI-010-02	Unit tests, SI-02	RA-302, RA-305
UTS-CSI-010-03	Unit tests, SI-03	RA-603, RA-605
UTS-CSI-010-10	Unit tests, SI-10	FW-605, FW-702, FW-704, FW-706, FW-803, FW-804
UTS-CSI-010-11	Unit tests, SI-11	FW-102, FW-201, FW-202, FW-203, FW-204, FW-206
UTS-CSI-010-12	Unit tests, SI-12	FW-102, FW-303, FW-304, FW-305, FW-306, FW-307, FW-308, FW-309
UTS-CSI-010-13	Unit tests, SI-13	FW-209, FW-210
UTS-CSI-010-14	Unit tests, SI-14	FW-401
UTS-CSI-010-15	Unit tests, SI-15	FW-604, FW-605
UTS-CSI-010-20	Unit tests, SI-20	AD-105
UTS-CSI-010-22	Unit tests, SI-22	AD-602
UTS-CSI-010-30	Unit tests, SI-30	DM-401, DM-402
UTS-CSI-010-31	Unit tests, SI-31	DM-103
UTS-CSI-010-32	Unit tests, SI-32	DM-202, DM-303
UTS-CSI-010-40	Unit tests, SI-40	CM-103
ITS-CSI-010 §1	Integration, SI-01	RA-1003, CM-305
ITS-CSI-010 §2	Integration, SI-02	RA-304, RA-307
ITS-CSI-010 §3	Integration, SI-03	RA-403, RA-404, RA-501, RA-502, RA-503, RA-505, RA-703, RA-705
ITS-CSI-010 §4	Integration, SI-04	RA-204, RA-205
ITS-CSI-010 §5	Integration, SI-05	RA-802, RA-805, RA-904
ITS-CSI-010 §6	Integration, SI-10	RA-204, RA-205, RA-505, RA-805, FW-101, FW-103, FW-104, FW-208, FW-302, FW-402, FW-501, FW-502, FW-601, FW-602, FW-603, FW-606, FW-607, FW-608, FW-701, FW-703, FW-705, FW-801, FW-802, FW-805, FW-806, FW-807, FW-808, AD-102
ITS-CSI-010 §7	Integration, SI-11	RA-802, FW-205, FW-207, FW-806
ITS-CSI-010 §8	Integration, SI-12	FW-301, FW-302, FW-310, FW-606, AD-502
ITS-CSI-010 §9	Integration, SI-13	RA-404, RA-501, FW-205, AD-501
ITS-CSI-010 §10	Integration, SI-14	FW-402, FW-403, AD-401, AD-403
ITS-CSI-010 §11	Integration, SI-15	FW-103
ITS-CSI-010 §12	Integration, SI-20	AD-102, AD-106
ITS-CSI-010 §13	Integration, SI-21	AD-401, AD-403, AD-501, AD-502
ITS-CSI-010 §15	Integration, SI-30	CM-202
ITS-CSI-010 §16	Integration, SI-31	RA-307, RA-705, RA-904, DM-504
ITS-CSI-010 §17	Integration, SI-32	DM-201
ITS-CSI-010 §19	Integration, SI-41	FW-801, FW-802, FW-805, FW-806, FW-807, FW-808
STS-CSI-010 §A	System test, Research Application	61 requirements: RA-101 to CM-401
STS-CSI-010 §B	System test, Firmware	52 requirements: FW-101 to CM-203
STS-CSI-010 §C	System test, Admin Application	40 requirements: AD-101 to AD-905

Test specification	Scope	Software requirements
STS-CSI-010 §D	System test, Data Management	23 requirements: DM-101 to CM-304
STS-CSI-010 §E	System test, Common	4 requirements: CM-101 to CM-201

7.3 Validation studies to user needs

Study	Title	User needs	Result
VAL-CSI-01	Method comparison	UN-01, UN-03	Pass
VAL-CSI-02	Clinical sort purity, recovery and viability	UN-02, UN-05, UN-15	Pass (after ANM-0056)
VAL-CSI-03	Summative usability (human factors validation)	UN-06, UN-09	Pass (after ANM-0060)
VAL-CSI-04	Multi-site precision	UN-08, UN-01	Pass
VAL-CSI-05	Laboratory workflow and throughput	UN-04	Pass
VAL-CSI-06	Installation in clinical laboratories	UN-07	Pass
VAL-CSI-07	Carryover with clinical specimens	UN-14	Pass
VAL-CSI-08	Temperature control with cells	UN-11	Pass
VAL-CSI-09	Records, review and release in use	UN-10	Pass (after ANM-0058)
VAL-CSI-10	Serviceability in the field	UN-12	Pass
VAL-CSI-11	Configuration and field upgrade	UN-16	Pass
VAL-CSI-12	Regulatory and labeling review	UN-13	Pass

7.4 Software items to requirements (design justified by requirements)

Item	Units (SAD)	Software requirements implemented
SI-01	5	RA-101, RA-102, RA-103, RA-104, RA-105, RA-504, RA-702, RA-1001, RA-1002, RA-1003, CM-301, CM-302, CM-305, CM-401
SI-02	5	RA-301, RA-302, RA-303, RA-304, RA-305, RA-306, RA-307, RA-901, RA-903, CM-303
SI-03	6	RA-401, RA-402, RA-403, RA-404, RA-405, RA-406, RA-407, RA-501, RA-502, RA-503, RA-505, RA-601, RA-602, RA-603, RA-604, RA-605, RA-701, RA-702, RA-703, RA-704, RA-705, RA-706
SI-04	4	RA-201, RA-202, RA-203, RA-204, RA-205, RA-206, RA-207, RA-208, RA-1002
SI-05	4	RA-701, RA-801, RA-802, RA-803, RA-804, RA-805, RA-806, RA-901, RA-902, RA-903, RA-904, AD-104, AD-203, AD-903
SI-10	10	RA-204, RA-205, RA-505, RA-805, FW-101, FW-103, FW-104, FW-208, FW-302, FW-402, FW-501, FW-502, FW-601, FW-602, FW-603, FW-605, FW-606, FW-607, FW-608, FW-701, FW-702, FW-703, FW-704, FW-705, FW-706, FW-801, FW-802, FW-803, FW-804, FW-805, FW-806, FW-807, FW-808, AD-102, CM-203, CM-303
SI-11	8	RA-802, FW-102, FW-201, FW-202, FW-203, FW-204, FW-205, FW-206, FW-207, FW-806
SI-12	6	FW-102, FW-301, FW-302, FW-303, FW-304, FW-305, FW-306, FW-307, FW-308, FW-309, FW-310, FW-606, AD-502, AD-504
SI-13	4	RA-404, RA-407, RA-501, FW-205, FW-209, FW-210, AD-501
SI-14	4	FW-401, FW-402, FW-403, AD-401, AD-403
SI-15	4	FW-103, FW-604, FW-605
SI-20	4	AD-101, AD-102, AD-103, AD-104, AD-105, AD-106, AD-107, AD-305
SI-21	6	AD-201, AD-202, AD-203, AD-204, AD-205, AD-301, AD-302, AD-303, AD-304, AD-305, AD-401, AD-402, AD-403, AD-404, AD-405, AD-501, AD-502, AD-503, AD-504, AD-505, AD-601, AD-901, AD-902, AD-903, AD-904, AD-905
SI-22	5	AD-601, AD-602, AD-603, AD-701, AD-702, AD-703, AD-801, AD-802
SI-30	5	DM-101, DM-102, DM-105, DM-203, DM-301, DM-304, DM-401, DM-402, DM-403, DM-404, DM-501, DM-502, DM-503, CM-202, CM-203, CM-304
SI-31	5	RA-307, RA-705, RA-706, RA-904, AD-902, DM-103, DM-104, DM-105, DM-504
SI-32	3	AD-103, AD-701, AD-802, DM-201, DM-202, DM-204, DM-302, DM-303, DM-304, DM-503
SI-40	4	RA-804, AD-801, DM-204, CM-101, CM-102, CM-103
SI-41	5	RA-303, FW-801, FW-802, FW-805, FW-806, FW-807, FW-808, CM-201, CM-302, CM-401

8. Anomaly Links

Every anomaly in ADL-CSI-001 is linked to the requirement or user need it affected. The result shown in the matrices reflects the final status after correction and regression testing.

Anomaly	Title	Requirements	Severity	Status
ANM-0003	Sort decision latency exceeds 20 μ s above 95,000 events/s	FW-204, SYS-461	2	Closed
ANM-0007	Safety supervisor removes HV enable during normal boot	FW-604, FW-103	3	Closed
ANM-0011	Clog response exceeds 2 s in 2 of 59 trials	SYS-543, FW-602	1	Closed
ANM-0012	PI CV 3.1% on DV-02 on day 2	SYS-401	2	Closed
ANM-0014	Breakoff control oscillates with 130 μ m nozzle	RA-501, SYS-613	3	Closed
ANM-0016	Laser calibration factor applied to wrong laser on shared beam spot	AD-202	2	Closed
ANM-0019	Carryover 0.7% with 1 mL microtube holder	SYS-471, RA-205	2	Closed
ANM-0021	Radiated immunity: event rate spikes at 3 V/m, 80 to 120 MHz	SYS-503	2	Closed
ANM-0023	After link loss, Research App shows Ready while firmware is in safe state	FW-805, RA-1002	2	Closed
ANM-0025	QC lockout window 25 h across daylight saving change	RA-803, SYS-622	2	Closed
ANM-0026	480/20 filter rule not enforced after field upgrade to 445 nm	AD-105	2	Closed
ANM-0028	Audit trail missing old value for gate edits in draft assay template	DM-201	2	Closed
ANM-0030	Replayed command accepted after reconnect (penetration test)	FW-802, FW-803	2	Closed
ANM-0032	Laser warning label missing on rear panel of 5-laser build	SYS-513	2	Closed
ANM-0034	PMT HV readback fault at 1,000 V	FW-304	3	Closed
ANM-0035	Internal log loses last entries on power loss	FW-703	3	Closed
ANM-0037	384-well plate edge wells off target	SYS-483, AD-504	3	Closed
ANM-0039	Sample chamber takes 52 min to reach 4 °C at 30 °C ambient	SYS-4101	3	Closed
ANM-0040	DV-05 needed realignment after ISTA 3E transport test	SYS-1201	2	Closed
ANM-0042	Idle session lock not applied during calibration wizard	CM-103	3	Closed
ANM-0044	Memory growth in plotting library after 20 h acquisition	CM-303	3	Closed
ANM-0045	FCS export missing \$PnV keyword for FSC	DM-401	3	Closed
ANM-0047	Test failure: purity 96.8% on DV-01	SYS-461	3	Closed - not a defect
ANM-0048	Duplicate of ANM-0023	RA-1002	2	Closed - duplicate
ANM-0051	Usability verification: clog recovery step skipped	SYS-902, RA-504	3	Closed
ANM-0053	German UI text truncated in sort layout editor	CM-305	4	Deferred

Anomaly	Title	Requirements	Severity	Status
ANM-0056	VAL-CSI-02 Site 3: CD34+ purity 93% on two specimens	UN-02	2	Closed
ANM-0058	VAL-CSI-09 Site 2: LIS export rejected by site LIS	UN-10, DM-404	3	Closed
ANM-0060	VAL-CSI-03: users unsure which nozzle is installed	UN-09	3	Closed
ANM-0062	Precision study: Site 1 day 3 QC warning on violet array	UN-08	3	Open - investigating
ANM-0064	Data Management search slow above 50,000 records	UN-10	4	Deferred
ANM-0066	Plate voltage readback drift warning after 6 h sorting	FW-307	3	Open - fix verified

9. Matrix E: Critical Characteristics to Production Controls

Design transfer extension (DTP-CSI-001 Section 6): each critical characteristic tested in the factory acceptance test traces from its design requirement to the production control in FTS-CSI-001.

Characteristic	Factory limit	Design requirement	Production control
Fluorescence resolution (PI)	$CV \leq 2.7\%$	SRS-CSI-001 §4.1	FTS-CSI-001 §5.1
Fluorescence resolution (Hoechst)	$CV \leq 3.2\%$	SRS-CSI-001 §4.1	FTS-CSI-001 §5.2
Linearity	1.97–2.03	SRS-CSI-001 §4.2	FTS-CSI-001 §5.3
Scatter sensitivity	Resolved with ND 1.0 removed	SRS-CSI-001 §4.3	FTS-CSI-001 §5.4
Acquisition rate	No data loss at 75,000 events/s	SRS-CSI-001 §4.6	FTS-CSI-001 §5.5
Performance QC baseline	Pass; baseline stored in DHR	SRS-SW-CSI-010 §6.6	FTS-CSI-001 §5.6
Breakoff stability	≤ 5 px drift; Gap within target +1	SSD-CSI-700 §3.3	FTS-CSI-001 §5.7
Drop delay	Within ± 0.25 drop of manual method	SSD-CSI-700 §3.4	FTS-CSI-001 §5.8
Sort purity / yield	$\geq 99\%$ / $\geq 85\%$	SRS-CSI-001 §4.7	FTS-CSI-001 §5.9
Plate deposition (option)	$\geq 98\%$ of wells with exactly one bead	SSD-CSI-800 §4	FTS-CSI-001 §5.10
Clog response	All four actions within 1 s	SRS-CSI-001 §5.5	FTS-CSI-001 §5.11
Containment (with aerosol option)	No particles detected outside the sort chamber	SRS-CSI-001 §5.6	FTS-CSI-001 §5.12
Evacuator flow	≥ 7.5 / ≥ 32 CFM	SSD-CSI-900 §3.2	FTS-CSI-001 §5.13
Sample chamber temperature	± 1 °C at 45 min	SSD-CSI-100 §7.2	FTS-CSI-001 §5.14
Fluidics pressure	84–96 PSI	SSD-CSI-100 §4.2	FTS-CSI-001 §5.15
Electrical safety	100% pass	SRS-CSI-001 §5.1	FTS-CSI-001 §5.16
Acoustic noise	≤ 60 dBA	SRS-CSI-001 §3.4	FTS-CSI-001 §5.17
E-stop and interlocks	100% pass	SRS-CSI-001 §5.4	FTS-CSI-001 §5.18

10. Reference Documents

Document no.	Rev	Title
SRS-CSI-001	D	System Requirements Specification
SRS-SW-CSI-010	E	Software Requirements Specification
SAD-SW-CSI-011	C	Software Architecture Description
VVP-CSI-001	A	System Verification and Validation Plan
PVR-CSI-001 / SVR-CSI-001	A / A	System Performance Verification Report / Safety, EMC and Laser Report
SVR-SW-CSI-010	A	Software Verification Report
DVAL-CSI-001	A	Design Validation Report
HA-CSI-001 / RMR-CSI-001	C / A	Hazard Analysis / Risk Management Report
ADL-CSI-001	Export 07	Anomaly and Defect Tracking Log
DTP-CSI-001 / FTS-CSI-001	B / A	Design Transfer Plan / Factory Alignment and Test Specification
VVR-CSI-001	A	V&V Summary Report